

Federated Single-Index Bandits

Cooperative learning when the reward function is unknown

Presented by:

Harshitha (23b1071) Dept. of CSE

Nischal (23b1024) Dept. of CSE

Course: CS6007: Multi-Agent Machine Learning

Instructor: Prof. Avishek Ghosh

IIT Bombay

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Contextual Bandits with Unknown Links

Standard contextual bandits assume a known relationship between context and reward:

$$\mathbb{E}[y \mid x] = \underbrace{\langle x, \theta_* \rangle}_{\text{linear bandit}} \quad \text{or} \quad \underbrace{f(\langle x, \theta_* \rangle)}_{\substack{\text{generalized linear} \\ f \text{ is known}}}$$

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Unknown and non-monotone reward profiles

In practical applications like pricing, traffic routing, and dose-response, the mapping f is rarely known beforehand and often non-monotone.

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Consequence of misspecification

When f is misspecified, standard methods like LinUCB or GLM-UCB suffer **linear regret** $\Omega(T)$ rather than sublinear growth.¹

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Single-Index Bandits (SIB)

System model

At round t , the learner observes K arms $\mathcal{X}_t = \{x_{t,a}\}_{a \in [K]}$ drawn i.i.d. from density p on $\mathbb{R}^d$, chooses x_t , and receives reward

$$y_t = f(\langle x_t, \theta_* \rangle) + \eta_t, \quad \eta_t \text{ } \sigma\text{-sub-Gaussian,}$$

where both the index direction $\theta_* \in \mathbb{R}^d$ and the link function $f : \mathbb{R} \rightarrow \mathbb{R}$ are unknown. We normalize $\|\theta_*\|_1 = 1$.



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Estimating θ_* via Stein's identity with score $S(x) = -\nabla \log p(x)$

$$\mathbb{E}[y S(x)] = \mathbb{E}[f'(\langle x, \theta_* \rangle)] \theta_* = \mu_* \theta_*$$

The expectation isolates the direction θ_* up to a scalar constant μ_* . This lets us estimate θ_* directly using empirical sample averages without fitting f .



Prior Literature and Positioning

	Unknown Link f	Multi- Agent	Safety Constraints	Optimal Regret Rate
Abbasi-Yadkori et al. '11 (OFUL)	✗	✗	✗	—
Chowdhury & Gopalan '17 (IGP-UCB)	✓	✗	✗	—
Dubey & Pentland '20 (Coop-KernelUCB)	✓	✓	✗	—
Amani & Thrampoulidis '21 (Safe-DLUCB)	✗	✓	✓	—
Kang et al. '26 (ESTOR / GSTOR)	✓	✗	✗	✗ $T^{3/4}$
Arya et al. '26 (kernel SIB)	✓	✗	✗	—
Dey, Bhore & Ghosh '26 (ZoomSIB-UCB)	✓	✗	✗	✓ $\tilde{\Theta}(T^{2/3})$
This project (Fed-ZoomSIB)	✓	✓	(ext.)	?



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Non-parametric links, multi-agent cooperation, and safety have each been studied separately.
 To our knowledge, single-index contextual bandits have not been studied in a collaborative or federated setting.

Problem Formulation: Multi-Agent SIB

Setting

Consider N decentralized agents on a connected communication graph G . At each round t , each agent i observes an independent arm set, pulls $x_{i,t}$, and observes $y_{i,t} = f(\langle x_{i,t}, \theta_* \rangle) + \eta_{i,t}$. The parameter θ_* and link f are identical across agents but unknown. Agents can exchange messages with their neighbors subject to a communication budget.

Objective: Cumulative Network Regret

$$R_T(N) = \sum_{t=1}^T \sum_{i=1}^N \left[f(\langle x_{i,t}^*, \theta_* \rangle) - f(\langle x_{i,t}, \theta_* \rangle) \right]$$

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Can decentralized collaboration improve per-agent regret when neither the index direction nor the link function is known?



Two Sharing Schemes in ZoomSIB-UCB

ZoomSIB-UCB (Dey, Bhore & Ghosh '26) operates in two sequential phases. Each phase provides a distinct point of collaboration:

Phase 1: Estimating θ_*

Each agent runs a truncated Stein estimator over T_0 initial rounds.

Because the estimator is a sample mean, averaging local estimates across agents produces the exact pooled centralized estimate in a single message (d floats), with no iterative consensus.

Phase 2: Binned UCB

The projected line $[-1, 1]$ is split into $N_{\text{bins}} = O(T^{1/3})$ intervals, running a sleeping UCB.

Each agent maintains two running scalars per bin (n_j, S_j) . Sharing these statistics across graph G corresponds to a cooperative multi-armed bandit with statistical pooling.

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Proposition 4.7 in Dey et al. proves a modular reduction: any valid sleeping-bandit subroutine can be used in Phase 2 while preserving regret guarantees. We exploit this by plugging in a cooperative sleeping bandit.



Core Research Questions

Rather than assuming theoretical gains upfront, we examine how the two phases contribute to speedup under bandwidth limits.

Questions under investigation

- 1 **Per-agent regret scaling:** Does collaboration yield the expected $O(1/\sqrt{N})$ per-agent regret reduction?
- 2 **Phase asymmetry:** Phase 1 pooling is algebraically exact, while Phase 2 pooling is statistical. Which phase drives the empirical performance gain?
- 3 **Communication budgets:** How far can message frequency be reduced (via periodic, event-triggered, or quantized exchanges) before regret degrades?
- 4 **Network topology:** How sensitive is performance to the graph spectral gap $1 - |\lambda_2(\mathbf{P})|$ across different dimensions d ?

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Potential extensions: (a) Safety constraints with unknown threshold functions $h(\langle x, \mu_* \rangle) \leq c$; (b) clustered index directions where agents belong to unknown latent groups.



Timeline and Evaluation Plan

Dates	Milestone	Deliverable
29 Sep – 10 Oct	Reproduce single-agent ZOOMSIB-UCB and baselines (Random, LinUCB, ESTOR, GSTOR, IGP-UCB)	Regret curves validating reproduction
11 – 22 Oct	Implement Fed-ZoomSIB: decentralized Stein averaging and cooperative sleeping UCB	Speedup curves; finite-time concentration bounds
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Evaluation metrics: Cumulative network regret, per-agent regret, message count, and total communication volume. We test sensitivity across network topologies, $N \in [5, 50]$, dimension d , and diverse link profiles on synthetic tasks as well as KDD Cup 99 and Forest Cover Type.



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Thank You!